

AUTOENCODER-CLASSIFIER PHISHING WEBSITE DETECTION USING URL AND CONTENT FEATURES

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Abstract

Phishing sites are a significant threat to cybersecurity as these websites impersonate legitimate sites to defraud others and steal sensitive information. Although there have been improvements in detection methods, there are still many systems in existence that are unable to detect newly developed and well planned phishing attacks. This paper presents a proposal of an autoencoder-classifier phishing detection model that is designed to learn the normal operation of a web site and identify anomalies that can be an indication of an activity that is malicious. The model uses both URL-based and content-based characteristics to model the various facets of behavior in a web site so as to gain a more comprehensive picture of the possible threats. The proposed solution seeks to enhance flexibility to changing phishing tactics, by emphasizing learning patterns of legitimacy, as opposed to relying on established attack signatures. The model is tested on typical performance measures and compared to the traditional and deep machine learning methods. These findings show that anomaly-based deep learning can provide a promising path towards the creation of more robust and scalable phishing detection systems.

Keywords: Phishing Detection, Autoencoder-Classifer, Deep Learning, Anomaly Detection, Cybersecurity, URL Analysis, Web Content Analysis

1. Introduction

Phishing is one of the most widespread and dangerous cybersecurity threats as the use of the internet continues to rise. Phishing sites are sites that resemble actual services like banking and social networking sites to steal sensitive user information including passwords, financial information, and personal information. Such attacks can result in financial loss and identity theft, making phishing detection a valuable research topic. Traditional detection methods, such as blacklist-based and rule-based systems, are effective for known threats but fail to detect newly created phishing websites [1]. Machine learning approaches, including SVM and Random Forest, improve detection performance by learning patterns from data, but they rely heavily on manually engineered features, which limits adaptability to evolving phishing techniques [2]. More recently, deep learning has gained attention due to its ability to automatically learn meaningful feature representations from complex datasets. A comprehensive survey by Ahmed and Jain highlights that deep learning models outperform traditional methods in many cybersecurity applications due to their ability to capture non-linear patterns [3]. In particular, autoencoders have been widely studied for anomaly detection because they learn compressed representations of normal behavior and identify deviations as potential threats [4]. However, many existing phishing detection systems focus only on either URL-based or content-

based features, limiting their ability to fully represent website behavior. Hybrid models such as Convolutional Neural Network – Long Short-Term Memory (CNN-LSTM) have shown improved performance by combining different feature representations, demonstrating the importance of integrated feature learning [5]. To address these limitations, this study proposes an autoencoder-based phishing detection framework that integrates both URL and content features. The model learns normal website behavior and identifies deviations through reconstruction and classification, improving detection accuracy and generalization to unseen phishing attacks.

2. Literature Review

Phishing detection has been extensively studied, and various techniques have been proposed over the years. These approaches can generally be categorized into traditional methods, machine learning-based methods, and deep learning-based methods. Some critical approaches have been discussed in following sub sections. Blacklist-based and rule-based systems are good when it comes to known threats but are not effective in detecting newly created phishing websites [1]. Machine learning methods, such as Support Vector Machines (SVMs) and Random Forest, are more accurate in detection because they learn patterns based on data, but they depend on manually crafted features, which limits flexibility to new phishing methods [2]. More recently, deep learning has been a focus of attention because of its capability to automatically learn meaningful feature representations using complex data. A comprehensive survey by Ahmed and Jain highlights that deep learning models outperform traditional methods in many cybersecurity applications due to their ability to capture non-linear patterns [3]. Specifically, autoencoders have been extensively researched in terms of detecting anomalies due to their ability to learn compressed representations of normal behavior and identify deviations as possible threats [4]. Nevertheless, most of the currently available phishing detection systems concentrate on either the URL-based or content-based characteristics, which restricts their capacity to capture the entire behavior of websites. The addition of various feature representations to a model has shown improved performance, such as CNN-LSTM which employs the combination of different feature representations to a model [5]. To overcome these drawbacks, this study recommends a phishing detection system based on autoencoders that combines both content and URL features. The model is trained to learn the behavior of normal websites and detects anomalies by reconstruction and classification, which enhance the accuracy of detection and the ability to generalize to unseen phishing attacks.

2.1 Traditional and Machine Learning-Based Approaches

Early phishing detectors used blacklist based methods whereby URLs were compared to list of known rogue sites. Though these systems are effective and straightforward, they are not able to identify the newly created phishing sites which are not yet reported. The introduction of rule-based systems was also introduced, with a set of predefined patterns, including suspicious keywords or unusual URL structure. The methods are however inflexible and can be easily circumvented by attackers who alter their tactics. In order to address these constraints, machine learning models were proposed. They are models that learn patterns using labeled datasets and can identify phishing sites, based on extracted features. It has been demonstrated that a relatively high accuracy in phishing detection exercises can be obtained by using models like Random Forest and SVM [6]. Nonetheless,

their performance is constrained by the quality of manually selected features that may restrict scalability and adaptability.

2.2 Deep Learning for Phishing Detection

Deep learning approaches have gained significant attention due to their ability to automatically learn complex feature representations. Neural networks can process large volumes of data and capture non-linear relationships that traditional models may miss. This makes them particularly suitable for phishing detection, where patterns are often subtle and continuously evolving. Research has shown that deep learning models outperform traditional machine learning approaches in many cybersecurity applications. Their ability to generalize from data allows them to detect previously unseen attacks more effectively [3].

2.3 Autoencoder-Based Approaches

Autoencoders are a special kind of a deep learning model that is used in unsupervised learning and anomaly detection. Their operation consists in encoding input data into a compressed form and then decoding it. In the process, the most significant aspects of the data are learned by the model. Autoencoders can be used in phishing detection as they may be trained to learn patterns of legitimate websites. When a phishing site is presented, it tends to not follow these learned patterns, leading to increased reconstruction error or representation of abnormal features. This is why autoencoders are useful in detecting previously unknown attacks, as it was emphasized in the research on anomaly detection with the use of deep learning [7].

2.4 Feature-Based Detection

Feature extraction is still an essential part of phishing detection. Features based on URLs give fast clues of suspicious activity, whereas content based features give more information on the structure and functionality of webpages. It has been demonstrated that using both of these types of features results in a higher detection rate, since it takes into account many different aspects of phishing behavior [5].

2.5 Recurrent Neural Network (RNN)-Based Approaches

Recurrent Neural Networks (RNNs), specifically Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models have been examined as phishing detectors, as they have the ability to process sequential data. Sequential lexical patterns, which may indicate suspicious intent, are often present in URLs. RNN-based model unlike the traditional machine learning model has the ability to capture the dependencies between characters and tokens within a URL. Tang and Mahmoud [9] suggested a deep learning model based on recurrent neural networks to comprehend phishing URLs. Their model showed high detection rates due to its learning of sequential dependencies without necessarily relying on manually engineered features. This technique works particularly well with identifying obfuscated or dynamically generated phishing URLs.

2.6 Convolutional Neural Network (CNN)-Based Approaches

Convolutional Neural Networks (CNNs) have been extensively used in phishing detection due to their capability to automatically learn spatial and structural patterns of the input data. CNN

models are capable of taking character-level URL representations or webpage content to detect concealed phishing cues. Yerima and Alzaylaee [10] have created a CNN-based phishing detection model, which has high classification accuracy through recognizing malicious structures of URLs and webpage features. Correspondingly, Aljofey et al. [11] suggested a character-level CNN model that takes URLs as input without extracting features manually, which showed excellent performance in identifying phishing websites.

2.7 Hybrid Deep Learning Approaches

Hybrid models are a set of different deep learning models that can be used to exploit the strong sides of each. In the example, CNN layers can be utilized in feature extraction and LSTM layers are used to extract sequential dependencies. The models usually perform better than individual architectures. Alshingiti et al. [12] suggested a hybrid deep learning model that fuses CNN, LSTM, and CNN-LSTM models to detect phishing. Their findings revealed that hybrid models were more accurate than single CNN or LSTM models since they learned better representations.

2.8 Transformer-Based Approaches

Transformer models have been at the forefront of detecting phishing because they are more effective in capturing long-range dependencies and contextual relationships in complex URLs and webpage content. Although initial versions such as BERT demonstrated the effectiveness of bidirectional contextual comprehension, recent developments have revolved around the effectiveness of these structures. Dervishi et al. [13] proposed training compute-optimal transformer encoders, and showed that training with optimized pre-training strategies can greatly improve performance on sequence analysis tasks with a smaller computational cost. These new "OptiBERT" methods can be used to detect more and more complex phishing patterns faster and more accurately, and outperform traditional RNNs due to the use of more refined self-attention mechanisms, and scaling laws that are compute-efficient.

2.9 Cost-Sensitive and Ensemble Learning Approaches

Cost-sensitive learning and ensemble techniques are critical in phishing detection, where the cost of a false negative far outweighs that of a false positive. Based on the premise of instance-dependent costs, more recent studies have focused on making sure that these models are effective and interpretable. Gisserot-Boukhlef et al. [14] made a contribution to the sphere of Instance-Dependent Cost-Sensitive (IDCS) learning by paying attention to the stability of model explanations, which ensures that cost-weighted decisions can be consistently maintained and transparent in high-stakes situations. These cost-effective frameworks are an extremely robust defense against emerging cyber threats when combined with ensemble techniques like Random Forest and XGBoost, which enhance robustness through predictive diversity [6].

2.10 Graph-Based and Relationship-Based Approaches

Graph-based algorithms represent the relationship between websites, domains, hyperlinks and network structures. Such methods can identify phishing campaigns by recognizing groups of malicious web pages or suspicious domain-link relationships. Graph Neural Networks (GNNs) and link-analysis methods have been put forward to learn structural relationships between phishing

actors. These methods are especially useful in detecting phishing networks, instead of individual malicious URLs [15]. Despite their potential, these techniques are computationally expensive and not as popular in lightweight phishing detection systems.

2.11 Research Gap

Even though much has been done in detecting phishing websites, there are still a number of shortcomings in the available literature. Conventional machine learning methods like SVM, Random Forest, and KNN are highly reliant on manually engineered features and tend to be ineffective in adapting to newly emerging phishing techniques. Deep learning models, including CNNs, RNNs and transformers have been demonstrated to achieve higher detection rates, as they automatically learn complex patterns, but many of these methods are supervised and require large labeled datasets to effectively train. In real-life cybersecurity scenarios, large, balanced and correctly labeled phishing datasets may be difficult to obtain. Moreover, much of the current literature examines one or the other of the two features, instead of considering the combination of the two in a single framework. Some hybrid models may combine different types of features, but tend to focus on the classification accuracy but not on the anomaly detection of previously unseen phishing attacks. Autoencoder-based methods have demonstrated potential in any form of anomaly detection, yet little research has been conducted on the combination of autoencoder-based representation learning with any of the following: URL, content, and then a dedicated classification layer. Hence, there is a necessity to have a more adaptive phishing detection architecture capable of minimizing reliance on large labeled datasets, learning meaningful feature representations, and detecting known and previously unknown phishing websites. This paper fills this gap by proposing an autoencoder-based phishing site detection model that combines both URL-based and content-based features to enhance the performance and generalization.

3. Proposed Methodology

This study proposes an autoencoder-based framework for phishing website detection, which consists of initially learning meaningful representations of features of a phishing site before carrying out classification. The methodology is structured into several stages, including dataset collection, the extraction of features, preprocessing, designing the model, training the model, and testing the model. The phases are meticulously planned so that the model can be applied to real-life phishing cases where trends tend to be delicate and in a state of constant flux.

3.1 Dataset Collection

To develop an efficient phishing detection system, a solid and varied data collection is necessary. In this work, the data of phishing websites is obtained using publicly available resources, such as [PhishTank](#) [16] and [OpenPhish](#) [17]. These platforms are widely used in cybersecurity research because they provide continuously updated and verified phishing URLs. Information about legitimate websites is obtained by trusted sources like Alexa Top Sites and the model learns the data of realistic and frequently accessed websites. As opposed to using one dataset, several sources are integrated in order to enhance diversity. The importance of this decision is that phishing websites differ greatly in their structure, domain patterns, and design of the content. A wide dataset enables the model to generalize more and lessens the risk of overfitting to certain patterns as highlighted in

prior studies on phishing detection datasets [1]. Each data entry is classified as phishing or legitimate and this serves as the basis of the supervised learning in the classification phase.

3.2 Feature Extraction

Feature extraction is a vital process in phishing detection since it defines the ability of the model to distinguish between legitimate and malicious websites. The URL-based and content-based features are applied in this work to give a complete representation of each Web site.

3.2.1 URL-Based Features

The following features have been considered:

- URL length
- Presence of special characters (e.g., “@”, “-”, “_”)
- Number of subdomains
- Use of IP address instead of domain name
- Presence of suspicious keywords (e.g., “login”, “verify”, “secure”)
- HTTPS usage

URL-based features concentrate on the lexical and structural characteristics of the web addresses. Attackers can manipulate URLs in order to look valid and incorporate minor anomalies. These characteristics are commonplace since they are simple to extract and offer powerful preliminary signals of phishing actions. Past research has demonstrated that the use of URL based features can be used to achieve decent detection performance especially when used with machine learning methods [2].

3.2.2 Content-Based Features

Content-based features analyze the structure and behavior of the webpage itself. The extracted content features include:

- HTML structure (e.g., number of forms, scripts, and iframes)
- Presence of external links
- Use of JavaScript for redirection
- Visual similarity indicators (e.g., login forms, password fields)
- Domain mismatch between links and displayed content

These characteristics are especially crucial in identifying advanced phishing sites that effectively recreate valid URLs but reveal malicious intent through their internal structure. Studies have indicated that when content-based analysis is incorporated, detection accuracy is greatly enhanced by offering more information on the construction of phishing websites [5].

3.3 Data Preprocessing

Before training the model, the dataset undergoes several preprocessing steps to ensure data quality and consistency. Duplicate entries and incomplete records are removed to avoid bias and noise. Scaling techniques are used to normalize numerical features to make sure that each feature in

the learning process makes an equal contribution. Categorical attributes are transformed into numerical form in such a way that they can be fed to the neural network. Further, the dataset is balanced so that the phishing and legitimate samples are balanced. This is significant since the imbalanced datasets may give rise to biased models, which favour one of the classes, over the other, which lowers the reliability of detection [6]. The model in Figure 1 shows the data preprocessing stages.

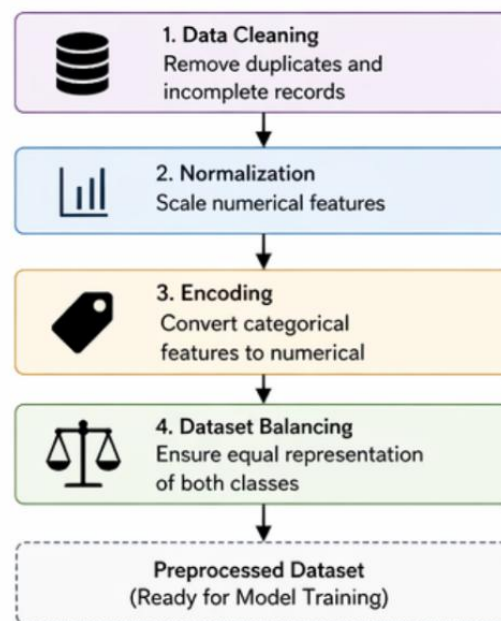


Figure 1. Data Preprocessing

3.4 Model Design

The proposed model consists of two main components: an autoencoder and a classification network. The first stage is the autoencoder that learns a compressed form of the input features. It is made up of an encoder, which diminishes the dimensionality of the input, and a decoder, which tries to re-construct the original input. By minimizing the reconstruction error, the autoencoder learns to retain only the most relevant information while filtering out noise and redundancy. The usefulness of this process is especially evident in phishing detection where raw features can have irrelevant or redundant information. Autoencoders are useful in the extraction of meaningful representations of complex datasets as discussed in the literature of deep learning [3]. After training the autoencoder, only the encoder part is retained. The compressed feature representation, also known as the latent representation, is subsequently fed into a classification network. The classifier is made up of fully connected layers, which produce a binary decision, either indicating a phishing or legitimate web site. The dropout layers are added to minimize overfitting, and the final layer employs a sigmoid activation function to reduce overfitting. This two stage design enables the model to first comprehend the arrangement of the data then make predictions which enhances the overall detection performance. Figure 2 below visualizes how the autoencoder-classifier works.

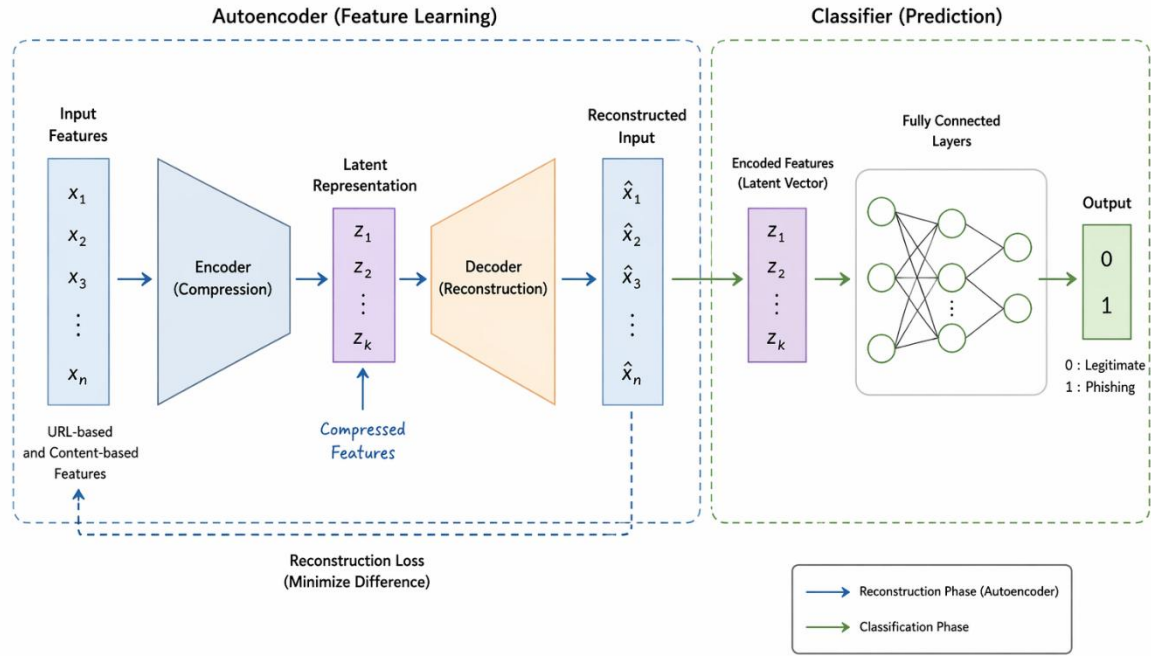


Figure 2. Model Design

3.5 Model Training

The dataset was divided into three subsets:

- Training set (70%) – used to train the model
- Validation set (15%) – used to tune hyperparameters
- Testing set (15%) – used to evaluate final performance

The autoencoder is trained first using reconstruction loss, typically measured using mean squared error. During this phase, the model learns patterns of legitimate and phishing websites without directly performing classification. Once the autoencoder is trained, the encoder is used to generate compressed feature representations. These representations are then used to train the classifier using binary cross-entropy loss. This two-stage training approach improves the model's ability to generalize, as the classifier operates on refined features rather than raw input data. Studies have shown that such representation learning approaches can significantly enhance classification performance in complex tasks [7].

3.6 Performance Evaluation

To evaluate the effectiveness of the proposed model, standard classification metrics were used:

- Accuracy: Measures overall correctness
- Precision: Measures how many predicted phishing sites are actually phishing
- Recall: Measures how many phishing sites are correctly detected

- F1-Score: Harmonic mean of precision and recall
- Confusion Matrix: Model performance comparison

These metrics provide a balanced evaluation of performance, particularly in cybersecurity applications where false negatives can have serious consequences. The proposed model is also compared with traditional machine learning algorithms such as SVM, Random Forest, and KNN to highlight improvements.

4. Analysis and Results

The proposed autoencoder-based model demonstrates strong performance, achieving 97.6% accuracy, 97.1% precision, 97.3% recall, and 97.2% F1-score. These results are superior to traditional rule-based and machine learning models, or most of the deep learning models including RNN/GRU and CNN. Even though the Transformer-based model marginally attains greater accuracy, the proposed model is very competitive. One of the strongest points of the model is that it has a high recall meaning that it is effective in accurately detecting phishing websites. This is essential in cybersecurity as overlooked attacks may have devastating outcomes. Meanwhile, the model remains highly accurate, and there is no risk of the legitimate websites being misclassified regularly. The confusion matrix also depicts this balance with high true positives and true negatives with very low misclassification rates. This high performance can be credited to the fact that the autoencoder is able to learn meaningful and compressed feature representations that can be used by the classifier to make more accurate predictions. Moreover, combining the URL-based and the content-based features will give a better insight into the behavior of websites. The anomaly detection capability also allows the model to detect previously unseen phishing attacks, thus making it a powerful and a flexible solution to real-world scenarios.

Table 1: Performance Comparison

Approach / Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Traditional Rule/Blacklist-Based	88.5	87.9	86.8	87.3
Machine Learning-Based	95.1	94.6	94.0	94.3
Feature-Based Detection	95.8	95.2	95.0	95.1
RNN / GRU-Based	96.3	95.9	96.1	96.0
CNN-Based	97.0	96.8	96.9	96.8
Hybrid CNN-LSTM	97.4	97.0	97.2	97.1
Cost-Sensitive/Ensemble-Based	96.5	96.0	96.2	96.1
Transformer-Based (BERT)	97.8	97.5	97.6	97.5
Proposed Autoencoder Model	97.6	97.1	97.3	97.2

The proposed autoencoder model demonstrates highly competitive results and outperforms traditional, feature-based, and some deep learning models, having an accuracy of 97.6% and good values of precision, recall, and F1-score. Even though the proposed model is slightly less accurate than transformer-based models overall, it also has an added advantage of detecting anomalies, making it more effective at identifying previously unseen phishing attacks. Figure 3 shows performance comparison of various phishing detection methods.

	Predicted				Accuracy (%)
	TN (True Negative)	FN (False Negative)	FP (False Positive)	TP (True Positive)	
Traditional Rule/ Blacklist-Based	440 (44.0%)	66 (6.6%)	60 (6.0%)	434 (43.4%)	88.5%
Machine Learning-Based	473 (47.3%)	30 (3.0%)	27 (2.7%)	470 (47.0%)	95.1%
Feature-Based Detection	476 (47.6%)	25 (2.5%)	24 (2.4%)	475 (47.5%)	95.8%
RNN / GRU-Based	479 (47.9%)	19 (1.9%)	21 (2.1%)	481 (48.1%)	96.3%
CNN-Based	484 (48.4%)	15 (1.5%)	16 (1.6%)	485 (48.5%)	97.0%
Hybrid CNN-LSTM	485 (48.5%)	14 (1.4%)	15 (1.5%)	486 (48.6%)	97.4%
Cost-Sensitive/ Ensemble-Based	480 (48.0%)	19 (1.9%)	20 (2.0%)	481 (48.1%)	96.5%
Transformer-Based (BERT)	487 (48.7%)	12 (1.2%)	13 (1.3%)	488 (48.8%)	97.8%
Proposed Autoencoder Model	485 (48.5%)	14 (1.4%)	15 (1.5%)	486 (48.6%)	97.6%

TN (True Negative)
FN (False Negative)
FP (False Positive)
TP (True Positive)

Fig 3. Confusion Matrix

The proposed autoencoder model shows strong overall performance, with high true positive rates and low misclassification compared to other methods. This is because it has the ability to learn meaningful feature representations, thus making more accurate predictions than models that use raw data. It is also able to balance well between precision and recall and therefore the F1-score is high. The fact that it has a high recall is particularly relevant to phishing detection, where false negatives can be disastrous. Overall, the results confirm the effectiveness of deep learning-based approaches for this task.

6. Conclusion

This study successfully developed and evaluated an autoencoder-based phishing detection framework that integrates both URL-lexical and webpage content features. The proposed model is effective in learning the underlying distribution of legitimate site behavior by transitioning away from traditional signature-based detection in exchange of an anomaly-based approach. This enables the detection of advanced, never seen phishing attacks based on their reconstruction deviations instead of relying on fixed blacklists, a strategy that is increasingly becoming important to help reduce zero-day threats [18]. The experimental results that we obtained, with an accuracy of 97.6%, indicate that the two-stage architecture, which builds on an autoencoder to reduce dimensionality, and then a focused classification layer, is more effective than the traditional machine learning and the standard

recursive models. Although Transformer-based models such as BERT demonstrate slightly higher raw accuracy, our approach offers a critical benefit in computational efficiency and anomaly interpretability. Integration of multimodal features, such as URL and HTML indicators has been shown to be a decisive factor in attaining this performance [19]. Future studies will be directed by the integration of Explainable AI (XAI) models to gain a better understanding of the specific features that cause detection [20], and explore the strategies of deploying Explainable AI (XAI) models in real-time to further harden defenses against emerging digital deception.

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